

4. Bud rate

Contents

1	Environment preparation	1
2	Data formatting	1
3	Model selection	2
3.1	Random effects selection	2
3.2	Fixed effects selection	3
3.3	Goodness-of-fit evaluation	5
3.4	Selected model summary	6
4	Estimated marginal means	23
4.1	Table	23
4.2	Plots	23
4.3	Common functions	23
4.4	Panel A: interaction between sleep condition and lineage	25
4.5	Panel B: lineage effect	25
4.6	Assembly	26

1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for generalized linear mixed-effects models (GLMM).

```

buds_df <-

  #' Load the raw data frame.
  df %>%

  #' Reshape the weekly bud counts from wide to long format.
  pivot_longer(cols = starts_with("buds_"),
               names_to = c(".value", "week"),
               names_sep = "_",) %>%

  #' Convert week and bud count to integers.
  mutate(week = as.integer(week),
         buds = as.integer(buds),
  #' Create a centered variable for the week.
         week_centered = week - mean(week, na.rm = TRUE)) %>%

  #' Remove weeks with missing bud count.
  filter(!is.na(buds)) %>%

  #' Keep only the variables needed for the analysis.
  select(sleep_condition, lineage, tum_state, week, week_centered,
         buds, id_unique, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

buds_df$lineage <- relevel(factor(buds_df$lineage), ref = "HO_MT")
buds_df$sleep_condition <- relevel(factor(buds_df$sleep_condition), ref = "Control")
buds_df$tum_state <- relevel(factor(buds_df$tum_state), ref = "Healthy")

```

3 Model selection

3.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

buds_pois_R_00 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition, REML = TRUE,
family = poisson, data = buds_df)
buds_pois_R_01 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_02 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition +
(week_centered | id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_03 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_04 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_05 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique) + (1 | parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_06 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique) + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_07 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique) + (1 | parent_unique) + (1 | replicate_unique), REML = TRUE, family = poisson,
data = buds_df)

```

```
AIC(buds_pois_R_00, buds_pois_R_01, buds_pois_R_02, buds_pois_R_03, buds_pois_R_04,
    buds_pois_R_05, buds_pois_R_06, buds_pois_R_07) %>% arrange(AIC)
```

```
##           df      AIC
## buds_pois_R_04 25 12602.13
## buds_pois_R_06 26 12604.13
## buds_pois_R_07 27 12606.13
## buds_pois_R_02 27 12616.80
## buds_pois_R_03 25 12625.02
## buds_pois_R_01 25 12625.68
## buds_pois_R_05 26 12626.02
## buds_pois_R_00 24 12628.14
```

The model with the replicate id as a random effect (buds_pois_R_04) has the lowest AIC, this random effect structure is therefore selected.

3.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```
buds_pois_F_00 <- glmmTMB(buds ~ 1 + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_01 <- glmmTMB(buds ~ week_centered + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_02 <- glmmTMB(buds ~ lineage + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_03 <- glmmTMB(buds ~ sleep_condition + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_04 <- glmmTMB(buds ~ tum_state + (1 | replicate_unique), REML = FALSE, family
= poisson, data = buds_df)
buds_pois_F_05 <- glmmTMB(buds ~ week_centered + lineage + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_06 <- glmmTMB(buds ~ week_centered + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_07 <- glmmTMB(buds ~ week_centered + tum_state + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_08 <- glmmTMB(buds ~ lineage + sleep_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_09 <- glmmTMB(buds ~ lineage + tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_10 <- glmmTMB(buds ~ sleep_condition + tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_11 <- glmmTMB(buds ~ week_centered * lineage + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_12 <- glmmTMB(buds ~ week_centered * sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_13 <- glmmTMB(buds ~ week_centered * tum_state + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_14 <- glmmTMB(buds ~ lineage * sleep_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
```

```

buds_pois_F_15 <- glmmTMB(buds ~ lineage * tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_16 <- glmmTMB(buds ~ sleep_condition * tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_17 <- glmmTMB(buds ~ week_centered + lineage + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_18 <- glmmTMB(buds ~ week_centered + lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_19 <- glmmTMB(buds ~ week_centered + sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_20 <- glmmTMB(buds ~ lineage + sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_21 <- glmmTMB(buds ~ week_centered * lineage + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_22 <- glmmTMB(buds ~ week_centered * sleep_condition + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_23 <- glmmTMB(buds ~ week_centered * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_24 <- glmmTMB(buds ~ week_centered * lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_25 <- glmmTMB(buds ~ week_centered * sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_26 <- glmmTMB(buds ~ week_centered * tum_state + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_27 <- glmmTMB(buds ~ lineage * sleep_condition + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_28 <- glmmTMB(buds ~ lineage * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_29 <- glmmTMB(buds ~ sleep_condition * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_30 <- glmmTMB(buds ~ lineage * sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_31 <- glmmTMB(buds ~ lineage * tum_state + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_32 <- glmmTMB(buds ~ sleep_condition * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_33 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_34 <- glmmTMB(buds ~ week_centered * lineage * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_35 <- glmmTMB(buds ~ week_centered * sleep_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_36 <- glmmTMB(buds ~ lineage * sleep_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)

AIC(buds_pois_F_00, buds_pois_F_01, buds_pois_F_02, buds_pois_F_03, buds_pois_F_04,
buds_pois_F_05, buds_pois_F_06, buds_pois_F_07, buds_pois_F_08, buds_pois_F_09,
buds_pois_F_10, buds_pois_F_11, buds_pois_F_12, buds_pois_F_13, buds_pois_F_14,
buds_pois_F_15, buds_pois_F_16, buds_pois_F_17, buds_pois_F_18, buds_pois_F_19,
buds_pois_F_20, buds_pois_F_21, buds_pois_F_22, buds_pois_F_23, buds_pois_F_24,
buds_pois_F_25, buds_pois_F_26, buds_pois_F_27, buds_pois_F_28, buds_pois_F_29,
buds_pois_F_30, buds_pois_F_31, buds_pois_F_32, buds_pois_F_33, buds_pois_F_34,
buds_pois_F_35, buds_pois_F_36) %>% arrange(AIC)

```

##		df	AIC
##	buds_pois_F_33	25	12489.87
##	buds_pois_F_21	14	12494.81
##	buds_pois_F_11	13	12500.31
##	buds_pois_F_24	14	12500.54
##	buds_pois_F_34	19	12509.36
##	buds_pois_F_27	14	12561.96
##	buds_pois_F_22	10	12563.02
##	buds_pois_F_17	9	12564.93
##	buds_pois_F_05	8	12570.68
##	buds_pois_F_18	9	12570.97
##	buds_pois_F_23	10	12572.75
##	buds_pois_F_28	11	12574.36
##	buds_pois_F_14	13	12862.38
##	buds_pois_F_30	14	12862.49
##	buds_pois_F_08	8	12866.84
##	buds_pois_F_32	10	12867.85
##	buds_pois_F_20	9	12868.47
##	buds_pois_F_31	11	12870.53
##	buds_pois_F_36	19	12871.30
##	buds_pois_F_02	7	12874.06
##	buds_pois_F_09	8	12874.72
##	buds_pois_F_15	10	12878.25
##	buds_pois_F_25	6	12881.20
##	buds_pois_F_29	6	12882.98
##	buds_pois_F_19	5	12883.40
##	buds_pois_F_35	9	12883.43
##	buds_pois_F_26	6	12885.35
##	buds_pois_F_07	4	12893.10
##	buds_pois_F_12	5	12894.87
##	buds_pois_F_13	5	12895.04
##	buds_pois_F_06	4	12896.94
##	buds_pois_F_01	3	12902.29
##	buds_pois_F_16	5	13173.44
##	buds_pois_F_10	4	13174.39
##	buds_pois_F_04	3	13186.02
##	buds_pois_F_03	3	13189.04
##	buds_pois_F_00	2	13195.82

The model with the interaction between the week, the lineage and the sleep condition has the lowest AIC and is therefore selected (buds_pois_F_33).

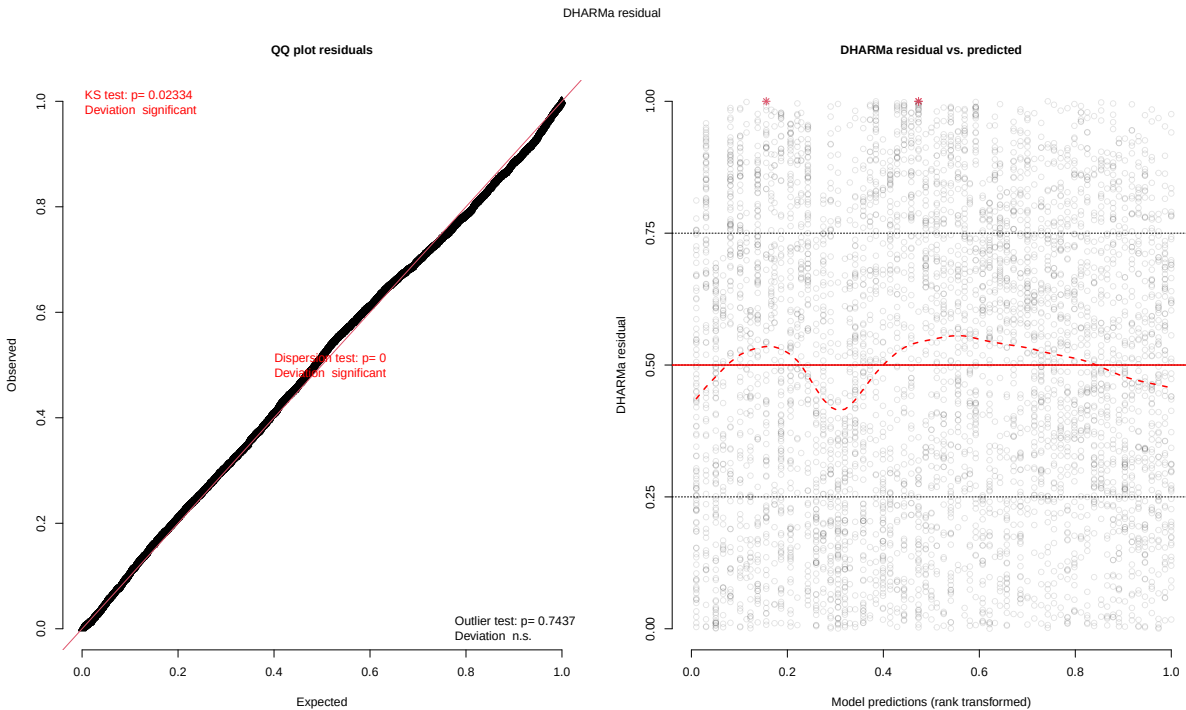
3.3 Goodness-of-fit evaluation

Refit the selected model.

```
buds_pois_mod_33 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
```

Assess goodness-of-fit of the selected model using simulated residuals.

```
buds_pois_mod_33_res <- simulateResiduals(buds_pois_mod_33, plot = TRUE)
```



3.4 Selected model summary

Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_MT")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
```

##	Estimate
## (Intercept)	0.457269
## week_centered	0.040455
## lineageHC_MT	-0.573663
## lineageHO_SPC	-0.256052
## lineageHO_SPT	-0.092013
## lineageHO_VLN	-1.013640
## lineageHV_GAL	-0.061518
## sleep_conditionMechanical sleep disruption	-0.063490
## week_centered:lineageHC_MT	0.227553
## week_centered:lineageHO_SPC	0.101016
## week_centered:lineageHO_SPT	0.126876
## week_centered:lineageHO_VLN	0.178343
## week_centered:lineageHV_GAL	0.140067
## week_centered:sleep_conditionMechanical sleep disruption	-0.007714
## lineageHC_MT:sleep_conditionMechanical sleep disruption	-0.209042
## lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.140629
## lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.030166
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	-0.119088
## lineageHV_GAL:sleep_conditionMechanical sleep disruption	-0.061672
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	-0.007118
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption	-0.065524
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.042134
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.034530
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption	-0.098026
##	Std. Error
## (Intercept)	0.044793
## week_centered	0.023019
## lineageHC_MT	0.086495
## lineageHO_SPC	0.066325
## lineageHO_SPT	0.066574
## lineageHO_VLN	0.074958
## lineageHV_GAL	0.064093
## sleep_conditionMechanical sleep disruption	0.057329
## week_centered:lineageHC_MT	0.044824
## week_centered:lineageHO_SPC	0.034484
## week_centered:lineageHO_SPT	0.034150
## week_centered:lineageHO_VLN	0.040251
## week_centered:lineageHV_GAL	0.033253
## week_centered:sleep_conditionMechanical sleep disruption	0.033359
## lineageHC_MT:sleep_conditionMechanical sleep disruption	0.125988
## lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.083929
## lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.084802
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.103689
## lineageHV_GAL:sleep_conditionMechanical sleep disruption	0.084267
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	0.069369
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.048808
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.048697
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.059922
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption	0.049032
##	z value
## (Intercept)	10.209
## week_centered	1.757
## lineageHC_MT	-6.632

```

## lineageHO_SPC -3.861
## lineageHO_SPT -1.382
## lineageHO_VLN -13.523
## lineageHV_GAL -0.960
## sleep_conditionMechanical sleep disruption -1.107
## week_centered:lineageHC_MT 5.077
## week_centered:lineageHO_SPC 2.929
## week_centered:lineageHO_SPT 3.715
## week_centered:lineageHO_VLN 4.431
## week_centered:lineageHV_GAL 4.212
## week_centered:sleep_conditionMechanical sleep disruption -0.231
## lineageHC_MT:sleep_conditionMechanical sleep disruption -1.659
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 1.676
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.356
## lineageHO_VLN:sleep_conditionMechanical sleep disruption -1.149
## lineageHV_GAL:sleep_conditionMechanical sleep disruption -0.732
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption -0.103
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption -1.343
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.865
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.576
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -1.999
## Pr(>|z|)
## (Intercept) < 2e-16
## week_centered 0.078845
## lineageHC_MT 3.30e-11
## lineageHO_SPC 0.000113
## lineageHO_SPT 0.166935
## lineageHO_VLN < 2e-16
## lineageHV_GAL 0.337142
## sleep_conditionMechanical sleep disruption 0.268097
## week_centered:lineageHC_MT 3.84e-07
## week_centered:lineageHO_SPC 0.003397
## week_centered:lineageHO_SPT 0.000203
## week_centered:lineageHO_VLN 9.39e-06
## week_centered:lineageHV_GAL 2.53e-05
## week_centered:sleep_conditionMechanical sleep disruption 0.817126
## lineageHC_MT:sleep_conditionMechanical sleep disruption 0.097072
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.093821
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.722053
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.250756
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.464253
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.918272
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.179432
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.386910
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.564451
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.045582
##
## (Intercept) ***
## week_centered .
## lineageHC_MT ***
## lineageHO_SPC ***
## lineageHO_SPT ***
## lineageHO_VLN ***
## lineageHV_GAL

```

```

## sleep_conditionMechanical sleep disruption
## week_centered:lineageHC_MT ***
## week_centered:lineageHO_SPC **
## week_centered:lineageHO_SPT ***
## week_centered:lineageHO_VLN ***
## week_centered:lineageHV_GAL ***
## week_centered:sleep_conditionMechanical sleep disruption
## lineageHC_MT:sleep_conditionMechanical sleep disruption .
## lineageHO_SPC:sleep_conditionMechanical sleep disruption .
## lineageHO_SPT:sleep_conditionMechanical sleep disruption
## lineageHO_VLN:sleep_conditionMechanical sleep disruption
## lineageHV_GAL:sleep_conditionMechanical sleep disruption
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPC")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) 0.20122
## week_centered 0.14147
## lineageHO_MT 0.25605
## lineageHC_MT -0.31761
## lineageHO_SPT 0.16404
## lineageHO_VLN -0.75759
## lineageHV_GAL 0.19453
## sleep_conditionMechanical sleep disruption 0.07714
## week_centered:lineageHO_MT -0.10102
## week_centered:lineageHC_MT 0.12654
## week_centered:lineageHO_SPT 0.02586
## week_centered:lineageHO_VLN 0.07733

```

## week_centered:lineageHV_GAL	0.03905
## week_centered:sleep_conditionMechanical sleep disruption	-0.07324
## lineageHO_MT:sleep_conditionMechanical sleep disruption	-0.14063
## lineageHC_MT:sleep_conditionMechanical sleep disruption	-0.34967
## lineageHO_SPT:sleep_conditionMechanical sleep disruption	-0.11046
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	-0.25972
## lineageHV_GAL:sleep_conditionMechanical sleep disruption	-0.20230
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption	0.06552
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	0.05841
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.10766
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.10005
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption	-0.03250
##	Std. Error
## (Intercept)	0.04895
## week_centered	0.02568
## lineageHO_MT	0.06633
## lineageHC_MT	0.08872
## lineageHO_SPT	0.06944
## lineageHO_VLN	0.07751
## lineageHV_GAL	0.06706
## sleep_conditionMechanical sleep disruption	0.06130
## week_centered:lineageHO_MT	0.03448
## week_centered:lineageHC_MT	0.04624
## week_centered:lineageHO_SPT	0.03599
## week_centered:lineageHO_VLN	0.04183
## week_centered:lineageHV_GAL	0.03514
## week_centered:sleep_conditionMechanical sleep disruption	0.03563
## lineageHO_MT:sleep_conditionMechanical sleep disruption	0.08393
## lineageHC_MT:sleep_conditionMechanical sleep disruption	0.12785
## lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.08753
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.10594
## lineageHV_GAL:sleep_conditionMechanical sleep disruption	0.08701
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption	0.04881
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	0.07049
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.05028
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.06121
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption	0.05060
##	z value
## (Intercept)	4.111
## week_centered	5.510
## lineageHO_MT	3.861
## lineageHC_MT	-3.580
## lineageHO_SPT	2.362
## lineageHO_VLN	-9.774
## lineageHV_GAL	2.901
## sleep_conditionMechanical sleep disruption	1.258
## week_centered:lineageHO_MT	-2.929
## week_centered:lineageHC_MT	2.736
## week_centered:lineageHO_SPT	0.718
## week_centered:lineageHO_VLN	1.849
## week_centered:lineageHV_GAL	1.111
## week_centered:sleep_conditionMechanical sleep disruption	-2.056
## lineageHO_MT:sleep_conditionMechanical sleep disruption	-1.676
## lineageHC_MT:sleep_conditionMechanical sleep disruption	-2.735

```

## lineageHO_SPT:sleep_conditionMechanical sleep disruption -1.262
## lineageHO_VLN:sleep_conditionMechanical sleep disruption -2.452
## lineageHV_GAL:sleep_conditionMechanical sleep disruption -2.325
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 1.343
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.829
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 2.141
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 1.634
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -0.642
## Pr(>|z|)
## (Intercept) 3.94e-05
## week_centered 3.59e-08
## lineageHO_MT 0.000113
## lineageHC_MT 0.000344
## lineageHO_SPT 0.018155
## lineageHO_VLN < 2e-16
## lineageHV_GAL 0.003721
## sleep_conditionMechanical sleep disruption 0.208241
## week_centered:lineageHO_MT 0.003397
## week_centered:lineageHC_MT 0.006213
## week_centered:lineageHO_SPT 0.472475
## week_centered:lineageHO_VLN 0.064496
## week_centered:lineageHV_GAL 0.266492
## week_centered:sleep_conditionMechanical sleep disruption 0.039815
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.093821
## lineageHC_MT:sleep_conditionMechanical sleep disruption 0.006238
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.206966
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.014220
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.020070
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.179432
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.407332
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.032252
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.102159
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.520692
##
## (Intercept) ***
## week_centered ***
## lineageHO_MT ***
## lineageHC_MT ***
## lineageHO_SPT *
## lineageHO_VLN ***
## lineageHV_GAL **
## sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_MT **
## week_centered:lineageHC_MT **
## week_centered:lineageHO_SPT
## week_centered:lineageHO_VLN .
## week_centered:lineageHV_GAL
## week_centered:sleep_conditionMechanical sleep disruption *
## lineageHO_MT:sleep_conditionMechanical sleep disruption .
## lineageHC_MT:sleep_conditionMechanical sleep disruption **
## lineageHO_SPT:sleep_conditionMechanical sleep disruption
## lineageHO_VLN:sleep_conditionMechanical sleep disruption *
## lineageHV_GAL:sleep_conditionMechanical sleep disruption *
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption

```

```
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption *
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPT")
summary(glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1      4728
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept)                                Estimate
## week_centered                                0.167331
## lineageHO_SPC                              -0.164039
## lineageHO_MT                                0.092013
## lineageHC_MT                               -0.481650
## lineageHO_VLN                              -0.921627
## lineageHV_GAL                               0.030494
## sleep_conditionMechanical sleep disruption -0.033324
## week_centered:lineageHO_SPC                 -0.025860
## week_centered:lineageHO_MT                 -0.126876
## week_centered:lineageHC_MT                  0.100677
## week_centered:lineageHO_VLN                 0.051467
## week_centered:lineageHV_GAL                 0.013190
## week_centered:sleep_conditionMechanical sleep disruption 0.034420
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.110464
## lineageHO_MT:sleep_conditionMechanical sleep disruption -0.030166
## lineageHC_MT:sleep_conditionMechanical sleep disruption -0.239207
## lineageHO_VLN:sleep_conditionMechanical sleep disruption -0.149254
## lineageHV_GAL:sleep_conditionMechanical sleep disruption -0.091837
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption -0.107659
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption -0.042134
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption -0.049252
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption -0.007605
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -0.140161
##
## Std. Error
```

## (Intercept)	0.049287
## week_centered	0.025225
## lineageHO_SPC	0.069436
## lineageHO_MT	0.066574
## lineageHC_MT	0.088903
## lineageHO_VLN	0.077724
## lineageHV_GAL	0.067302
## sleep_conditionMechanical sleep disruption	0.062486
## week_centered:lineageHO_SPC	0.035994
## week_centered:lineageHO_MT	0.034150
## week_centered:lineageHC_MT	0.045995
## week_centered:lineageHO_VLN	0.041552
## week_centered:lineageHV_GAL	0.034816
## week_centered:sleep_conditionMechanical sleep disruption	0.035476
## lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.087534
## lineageHO_MT:sleep_conditionMechanical sleep disruption	0.084802
## lineageHC_MT:sleep_conditionMechanical sleep disruption	0.128442
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.106626
## lineageHV_GAL:sleep_conditionMechanical sleep disruption	0.087829
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.050278
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption	0.048697
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	0.070412
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.061126
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption	0.050497
##	z value
## (Intercept)	7.411
## week_centered	6.633
## lineageHO_SPC	-2.362
## lineageHO_MT	1.382
## lineageHC_MT	-5.418
## lineageHO_VLN	-11.858
## lineageHV_GAL	0.453
## sleep_conditionMechanical sleep disruption	-0.533
## week_centered:lineageHO_SPC	-0.718
## week_centered:lineageHO_MT	-3.715
## week_centered:lineageHC_MT	2.189
## week_centered:lineageHO_VLN	1.239
## week_centered:lineageHV_GAL	0.379
## week_centered:sleep_conditionMechanical sleep disruption	0.970
## lineageHO_SPC:sleep_conditionMechanical sleep disruption	1.262
## lineageHO_MT:sleep_conditionMechanical sleep disruption	-0.356
## lineageHC_MT:sleep_conditionMechanical sleep disruption	-1.862
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	-1.400
## lineageHV_GAL:sleep_conditionMechanical sleep disruption	-1.046
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption	-2.141
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption	-0.865
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	-0.699
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	-0.124
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption	-2.776
##	Pr(> z)
## (Intercept)	1.26e-13
## week_centered	3.28e-11
## lineageHO_SPC	0.018155
## lineageHO_MT	0.166935

```

## lineageHC_MT 6.04e-08
## lineageHO_VLN < 2e-16
## lineageHV_GAL 0.650481
## sleep_conditionMechanical sleep disruption 0.593822
## week_centered:lineageHO_SPC 0.472475
## week_centered:lineageHO_MT 0.000203
## week_centered:lineageHC_MT 0.028607
## week_centered:lineageHO_VLN 0.215487
## week_centered:lineageHV_GAL 0.704790
## week_centered:sleep_conditionMechanical sleep disruption 0.331927
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.206966
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.722053
## lineageHC_MT:sleep_conditionMechanical sleep disruption 0.062549
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.161578
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.295727
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.032252
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.386910
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.484246
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.900992
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.005510
##
## (Intercept) ***
## week_centered ***
## lineageHO_SPC *
## lineageHO_MT
## lineageHC_MT ***
## lineageHO_VLN ***
## lineageHV_GAL
## sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPC
## week_centered:lineageHO_MT ***
## week_centered:lineageHC_MT *
## week_centered:lineageHO_VLN
## week_centered:lineageHV_GAL
## week_centered:sleep_conditionMechanical sleep disruption
## lineageHO_SPC:sleep_conditionMechanical sleep disruption
## lineageHO_MT:sleep_conditionMechanical sleep disruption
## lineageHC_MT:sleep_conditionMechanical sleep disruption .
## lineageHO_VLN:sleep_conditionMechanical sleep disruption
## lineageHV_GAL:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption *
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_VLN")
summary(glmnTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```
## Family: poisson ( log )
```

```

## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##           AIC           BIC      logLik -2*log(L)  df.resid
##    12602.1    12763.8    -6276.1    12552.1      4728
##
## Random effects:
##
## Conditional model:
##   Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate
## (Intercept)                        -0.556370
## week_centered                       0.218798
## lineageHO_SPT                      0.921627
## lineageHO_SPC                      0.757588
## lineageHO_MT                      1.013640
## lineageHC_MT                      0.439977
## lineageHV_GAL                     0.952121
## sleep_conditionMechanical sleep disruption -0.182578
## week_centered:lineageHO_SPT        -0.051467
## week_centered:lineageHO_SPC        -0.077327
## week_centered:lineageHO_MT        -0.178343
## week_centered:lineageHC_MT         0.049210
## week_centered:lineageHV_GAL       -0.038276
## week_centered:sleep_conditionMechanical sleep disruption 0.026816
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.149254
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.259718
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.119088
## lineageHC_MT:sleep_conditionMechanical sleep disruption -0.089953
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.057417
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.007605
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption -0.100054
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption -0.034530
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption -0.041648
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -0.132556
##                                     Std. Error
## (Intercept)                        0.060131
## week_centered                      0.033019
## lineageHO_SPT                      0.077724
## lineageHO_SPC                      0.077511
## lineageHO_MT                      0.074958
## lineageHC_MT                      0.095343
## lineageHV_GAL                     0.075608
## sleep_conditionMechanical sleep disruption 0.086399
## week_centered:lineageHO_SPT        0.041552
## week_centered:lineageHO_SPC        0.041827
## week_centered:lineageHO_MT        0.040251
## week_centered:lineageHC_MT        0.050688
## week_centered:lineageHV_GAL       0.040814

```

```

## week_centered:sleep_conditionMechanical sleep disruption 0.049778
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.106626
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.105936
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.103689
## lineageHC_MT:sleep_conditionMechanical sleep disruption 0.141613
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.106192
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.061126
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.061215
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.059922
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.078592
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.061383
## z value
## (Intercept) -9.253
## week_centered 6.626
## lineageHO_SPT 11.858
## lineageHO_SPC 9.774
## lineageHO_MT 13.523
## lineageHC_MT 4.615
## lineageHV_GAL 12.593
## sleep_conditionMechanical sleep disruption -2.113
## week_centered:lineageHO_SPT -1.239
## week_centered:lineageHO_SPC -1.849
## week_centered:lineageHO_MT -4.431
## week_centered:lineageHC_MT 0.971
## week_centered:lineageHV_GAL -0.938
## week_centered:sleep_conditionMechanical sleep disruption 0.539
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 1.400
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 2.452
## lineageHO_MT:sleep_conditionMechanical sleep disruption 1.149
## lineageHC_MT:sleep_conditionMechanical sleep disruption -0.635
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.541
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.124
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption -1.634
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption -0.576
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption -0.530
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -2.159
## Pr(>|z|)
## (Intercept) < 2e-16
## week_centered 3.44e-11
## lineageHO_SPT < 2e-16
## lineageHO_SPC < 2e-16
## lineageHO_MT < 2e-16
## lineageHC_MT 3.94e-06
## lineageHV_GAL < 2e-16
## sleep_conditionMechanical sleep disruption 0.0346
## week_centered:lineageHO_SPT 0.2155
## week_centered:lineageHO_SPC 0.0645
## week_centered:lineageHO_MT 9.39e-06
## week_centered:lineageHC_MT 0.3316
## week_centered:lineageHV_GAL 0.3483
## week_centered:sleep_conditionMechanical sleep disruption 0.5901
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.1616
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.0142
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.2508

```

```

## lineageHC_MT:sleep_conditionMechanical sleep disruption 0.5253
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.5887
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.9010
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.1022
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.5645
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.5962
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.0308
##
## (Intercept) ***
## week_centered ***
## lineageHO_SPT ***
## lineageHO_SPC ***
## lineageHO_MT ***
## lineageHC_MT ***
## lineageHV_GAL ***
## sleep_conditionMechanical sleep disruption *
## week_centered:lineageHO_SPT
## week_centered:lineageHO_SPC .
## week_centered:lineageHO_MT ***
## week_centered:lineageHC_MT
## week_centered:lineageHV_GAL
## week_centered:sleep_conditionMechanical sleep disruption
## lineageHO_SPT:sleep_conditionMechanical sleep disruption
## lineageHO_SPC:sleep_conditionMechanical sleep disruption *
## lineageHO_MT:sleep_conditionMechanical sleep disruption
## lineageHC_MT:sleep_conditionMechanical sleep disruption
## lineageHV_GAL:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HC_MT")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups          Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428

```

```

##
## Conditional model:
##
## (Intercept) -0.116393
## week_centered 0.268008
## lineageHO_VLN -0.439977
## lineageHO_SPT 0.481650
## lineageHO_SPC 0.317611
## lineageHO_MT 0.573663
## lineageHV_GAL 0.512144
## sleep_conditionMechanical sleep disruption -0.272531
## week_centered:lineageHO_VLN -0.049210
## week_centered:lineageHO_SPT -0.100677
## week_centered:lineageHO_SPC -0.126537
## week_centered:lineageHO_MT -0.227553
## week_centered:lineageHV_GAL -0.087487
## week_centered:sleep_conditionMechanical sleep disruption -0.014832
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.089953
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.239207
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.349671
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.209042
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.147370
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.041648
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.049252
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption -0.058406
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.007118
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -0.090908
##
## Std. Error
## (Intercept) 0.074015
## week_centered 0.038461
## lineageHO_VLN 0.095343
## lineageHO_SPT 0.088903
## lineageHO_SPC 0.088717
## lineageHO_MT 0.086495
## lineageHV_GAL 0.087060
## sleep_conditionMechanical sleep disruption 0.112202
## week_centered:lineageHO_VLN 0.050688
## week_centered:lineageHO_SPT 0.045995
## week_centered:lineageHO_SPC 0.046244
## week_centered:lineageHO_MT 0.044824
## week_centered:lineageHV_GAL 0.045329
## week_centered:sleep_conditionMechanical sleep disruption 0.060821
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.141613
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.128442
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.127850
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.125988
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.128148
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.078592
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.070412
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.070488
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.069369
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.070639
##
## z value
## (Intercept) -1.573

```

```

## week_centered 6.968
## lineageHO_VLN -4.615
## lineageHO_SPT 5.418
## lineageHO_SPC 3.580
## lineageHO_MT 6.632
## lineageHV_GAL 5.883
## sleep_conditionMechanical sleep disruption -2.429
## week_centered:lineageHO_VLN -0.971
## week_centered:lineageHO_SPT -2.189
## week_centered:lineageHO_SPC -2.736
## week_centered:lineageHO_MT -5.077
## week_centered:lineageHV_GAL -1.930
## week_centered:sleep_conditionMechanical sleep disruption -0.244
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.635
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 1.862
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 2.735
## lineageHO_MT:sleep_conditionMechanical sleep disruption 1.659
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 1.150
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.530
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.699
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption -0.829
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.103
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption -1.287
## Pr(>|z|)
## (Intercept) 0.115820
## week_centered 3.21e-12
## lineageHO_VLN 3.94e-06
## lineageHO_SPT 6.04e-08
## lineageHO_SPC 0.000344
## lineageHO_MT 3.30e-11
## lineageHV_GAL 4.04e-09
## sleep_conditionMechanical sleep disruption 0.015143
## week_centered:lineageHO_VLN 0.331626
## week_centered:lineageHO_SPT 0.028607
## week_centered:lineageHO_SPC 0.006213
## week_centered:lineageHO_MT 3.84e-07
## week_centered:lineageHV_GAL 0.053602
## week_centered:sleep_conditionMechanical sleep disruption 0.807336
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.525295
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.062549
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.006238
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.097072
## lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.250145
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.596166
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.484246
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.407332
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.918272
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption 0.198115
##
## (Intercept)
## week_centered ***
## lineageHO_VLN ***
## lineageHO_SPT ***
## lineageHO_SPC ***

```

```

## lineageHO_MT ***
## lineageHV_GAL ***
## sleep_conditionMechanical sleep disruption *
## week_centered:lineageHO_VLN
## week_centered:lineageHO_SPT *
## week_centered:lineageHO_SPC **
## week_centered:lineageHO_MT ***
## week_centered:lineageHV_GAL .
## week_centered:sleep_conditionMechanical sleep disruption
## lineageHO_VLN:sleep_conditionMechanical sleep disruption
## lineageHO_SPT:sleep_conditionMechanical sleep disruption .
## lineageHO_SPC:sleep_conditionMechanical sleep disruption **
## lineageHO_MT:sleep_conditionMechanical sleep disruption .
## lineageHV_GAL:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHV_GAL:sleep_conditionMechanical sleep disruption
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HV_GAL")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) Estimate
## week_centered 0.18052
## lineageHC_MT -0.51214
## lineageHO_VLN -0.95212
## lineageHO_SPT -0.03049
## lineageHO_SPC -0.19453
## lineageHO_MT 0.06152
## sleep_conditionMechanical sleep disruption -0.12516
## week_centered:lineageHC_MT 0.08749
## week_centered:lineageHO_VLN 0.03828

```

## week_centered:lineageHO_SPT	-0.01319
## week_centered:lineageHO_SPC	-0.03905
## week_centered:lineageHO_MT	-0.14007
## week_centered:sleep_conditionMechanical sleep disruption	-0.10574
## lineageHC_MT:sleep_conditionMechanical sleep disruption	-0.14737
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	-0.05742
## lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.09184
## lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.20230
## lineageHO_MT:sleep_conditionMechanical sleep disruption	0.06167
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	0.09091
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.13256
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.14016
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.03250
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption	0.09803
##	Std. Error
## (Intercept)	0.04592
## week_centered	0.02400
## lineageHC_MT	0.08706
## lineageHO_VLN	0.07561
## lineageHO_SPT	0.06730
## lineageHO_SPC	0.06706
## lineageHO_MT	0.06409
## sleep_conditionMechanical sleep disruption	0.06174
## week_centered:lineageHC_MT	0.04533
## week_centered:lineageHO_VLN	0.04081
## week_centered:lineageHO_SPT	0.03482
## week_centered:lineageHO_SPC	0.03514
## week_centered:lineageHO_MT	0.03325
## week_centered:sleep_conditionMechanical sleep disruption	0.03593
## lineageHC_MT:sleep_conditionMechanical sleep disruption	0.12815
## lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.10619
## lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.08783
## lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.08701
## lineageHO_MT:sleep_conditionMechanical sleep disruption	0.08427
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption	0.07064
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption	0.06138
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption	0.05050
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption	0.05060
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption	0.04903
##	z value
## (Intercept)	8.618
## week_centered	7.522
## lineageHC_MT	-5.883
## lineageHO_VLN	-12.593
## lineageHO_SPT	-0.453
## lineageHO_SPC	-2.901
## lineageHO_MT	0.960
## sleep_conditionMechanical sleep disruption	-2.027
## week_centered:lineageHC_MT	1.930
## week_centered:lineageHO_VLN	0.938
## week_centered:lineageHO_SPT	-0.379
## week_centered:lineageHO_SPC	-1.111
## week_centered:lineageHO_MT	-4.212
## week_centered:sleep_conditionMechanical sleep disruption	-2.943

```

## lineageHC_MT:sleep_conditionMechanical sleep disruption -1.150
## lineageHO_VLN:sleep_conditionMechanical sleep disruption -0.541
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 1.046
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 2.325
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.732
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 1.287
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 2.159
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 2.776
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.642
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 1.999
## Pr(>|z|)
## (Intercept) < 2e-16
## week_centered 5.38e-14
## lineageHC_MT 4.04e-09
## lineageHO_VLN < 2e-16
## lineageHO_SPT 0.65048
## lineageHO_SPC 0.00372
## lineageHO_MT 0.33714
## sleep_conditionMechanical sleep disruption 0.04265
## week_centered:lineageHC_MT 0.05360
## week_centered:lineageHO_VLN 0.34833
## week_centered:lineageHO_SPT 0.70479
## week_centered:lineageHO_SPC 0.26649
## week_centered:lineageHO_MT 2.53e-05
## week_centered:sleep_conditionMechanical sleep disruption 0.00325
## lineageHC_MT:sleep_conditionMechanical sleep disruption 0.25015
## lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.58872
## lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.29573
## lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.02007
## lineageHO_MT:sleep_conditionMechanical sleep disruption 0.46425
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption 0.19812
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption 0.03081
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption 0.00551
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption 0.52069
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption 0.04558
##
## (Intercept) ***
## week_centered ***
## lineageHC_MT ***
## lineageHO_VLN ***
## lineageHO_SPT **
## lineageHO_SPC **
## lineageHO_MT *
## sleep_conditionMechanical sleep disruption .
## week_centered:lineageHC_MT .
## week_centered:lineageHO_VLN
## week_centered:lineageHO_SPT
## week_centered:lineageHO_SPC
## week_centered:lineageHO_MT ***
## week_centered:sleep_conditionMechanical sleep disruption **
## lineageHC_MT:sleep_conditionMechanical sleep disruption
## lineageHO_VLN:sleep_conditionMechanical sleep disruption
## lineageHO_SPT:sleep_conditionMechanical sleep disruption
## lineageHO_SPC:sleep_conditionMechanical sleep disruption *

```

```
## lineageHO_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHC_MT:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_VLN:sleep_conditionMechanical sleep disruption *
## week_centered:lineageHO_SPT:sleep_conditionMechanical sleep disruption **
## week_centered:lineageHO_SPC:sleep_conditionMechanical sleep disruption
## week_centered:lineageHO_MT:sleep_conditionMechanical sleep disruption *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4 Estimated marginal means

4.1 Table

Create a character string containing the 6 exact centered values.

```
week_values <- 1:6 - mean(buds_df$week, na.rm = TRUE)
week_terms <- paste0("week_centered [", paste(week_values, collapse = ","), "]")
```

Compute the estimated marginal means of weekly bud production for every combination of week, sleep condition, and lineage, based on the selected model.

```
buds_emmeans <-
  as.data.frame(ggemmeans(buds_pois_mod_33,
    terms = c(week_terms, "sleep_condition", "lineage"),
    bias_correction = TRUE)) %>%
  arrange(facet, group, x)
```

Back-transform the centered week to the original scale (weeks 1 to 6) for plotting.

```
buds_emmeans$x <- buds_emmeans$x + mean(buds_df$week, na.rm = TRUE)
```

Round the values to ensure ggplot perfectly recognizes the integers 1 to 6.

```
buds_emmeans$x <- round(buds_emmeans$x)
```

4.2 Plots

4.3 Common functions

Function to plot, for a single lineage, the predicted weekly bud production by sleep condition.

```
buds_plot_treatment <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
  show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)

  prefix <- sub("_.*", "", lineage)
  prefix_format <- paste0(substr(prefix, 1, 1), tolower(substr(prefix, 2,
nchar(prefix))))
```

```

label_expr <- parse(text = paste0("bolditalic('", prefix_format, "') * bold('",
sub(".*_", "_", lineage), "'"))

ggplot(data_sub, aes(x = x, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 0.8) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  labs(x = if(show_x_title) "Age (weeks)" else NULL,
       y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
       else NULL,
       title = NULL) +
  theme_bw() +
  theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
        element_blank(),
        axis.ticks.x = if(show_x_text) element_line() else element_blank(),
        axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
        element_blank(),
        axis.ticks.y = if(show_y_text) element_line() else element_blank(),
        axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
        10)),
        axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
        margin = margin(l = 25)),
        axis.line.y.right = element_blank(),
        plot.margin = margin(l = 5, r = 5, t = 5, b = 5),
        legend.position = "none") +
  scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
  scale_y_continuous(position = "right") +
  coord_cartesian(ylim = c(0, 3.5)) +
  scale_color_manual(values = sleep_condition_colors) +
  scale_fill_manual(values = sleep_condition_colors) +
  annotate("text", x = 1.2, y = 3.3, label = label_expr,
         color = "black", size = 5, hjust = 0, vjust = 1) }

```

Function to plot, for a single lineage, the predicted weekly bud production pooled across sleep conditions.

```

buds_plot_lineage <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)
  data_lineage <- aggregate(cbind(predicted, conf.low, conf.high) ~ x, data = data_sub,
FUN = mean)
  data_lineage <- data_lineage[order(data_lineage$x), ]

  prefix <- sub(".*_", "", lineage)
  prefix_format <- paste0(substr(prefix, 1, 1), tolower(substr(prefix, 2,
nchar(prefix))))
  label_expr <- parse(text = paste0("bolditalic('", prefix_format, "') * bold('",
sub(".*_", "_", lineage), "'"))

  ggplot(data_lineage, aes(x = x, y = predicted)) +
    geom_line(linewidth = 0.8, color = "black") +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, fill = "black",
    linewidth = 0.3) +
    labs(x = if(show_x_title) "Age (weeks)" else NULL,
         y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
         else NULL,

```

```

    title = NULL) +
  theme_bw() +
  theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
  element_blank(),
        axis.ticks.x = if(show_x_text) element_line() else element_blank(),
        axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
  element_blank(),
        axis.ticks.y = if(show_y_text) element_line() else element_blank(),
        axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
  10)),
        axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
  margin = margin(l = 25)),
        axis.line.y.right = element_blank(),
        plot.margin = margin(l = 5, r = 5, t = 5, b = 5)) +
  scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
  scale_y_continuous(position = "right") +
  coord_cartesian(ylim = c(0, 3.5)) +
  annotate("text", x = 1.2, y = 3.3, label = label_expr,
          color = "black", size = 5, hjust = 0, vjust = 1) }

```

4.4 Panel A: interaction between sleep condition and lineage

```

A1 <- buds_plot_treatment("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
A2 <- buds_plot_treatment("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
A3 <- buds_plot_treatment("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
A4 <- buds_plot_treatment("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
A5 <- buds_plot_treatment("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title
= TRUE)
A6 <- buds_plot_treatment("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title
= TRUE)

```

Add a letter tag (A) directly to the first sub-plot before combining.

```

A1 <- A1 + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face = "bold"))

```

Combine A1 to A6 into panel A.

```

A <- (A1 + A2 + A3) / (A4 + A5 + A6) + plot_layout(heights = c(1, 1))

```

4.5 Panel B: lineage effect

```

B1 <- buds_plot_lineage("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
B2 <- buds_plot_lineage("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
B3 <- buds_plot_lineage("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
B4 <- buds_plot_lineage("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
B5 <- buds_plot_lineage("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title =
TRUE)
B6 <- buds_plot_lineage("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title =
TRUE)

```

Add a letter tag (B) directly to the first sub-plot before combining.

```
B1 <- B1 + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face = "bold"))
```

Combine B1 to B6 into panel B.

```
B <- (B1 + B2 + B3) / (B4 + B5 + B6) + plot_layout(heights = c(1, 1))
```

4.6 Assembly

Extract legend from the plot A6 to use later on the full figure.

```
legend_treatment <- cowplot::get_legend(  
  A6 +  
  labs(color = NULL, fill = NULL) +  
  theme(legend.position = "bottom",  
        legend.direction = "horizontal",  
        legend.text = element_text(size = 13)))
```

Build title graphical objects for each panel.

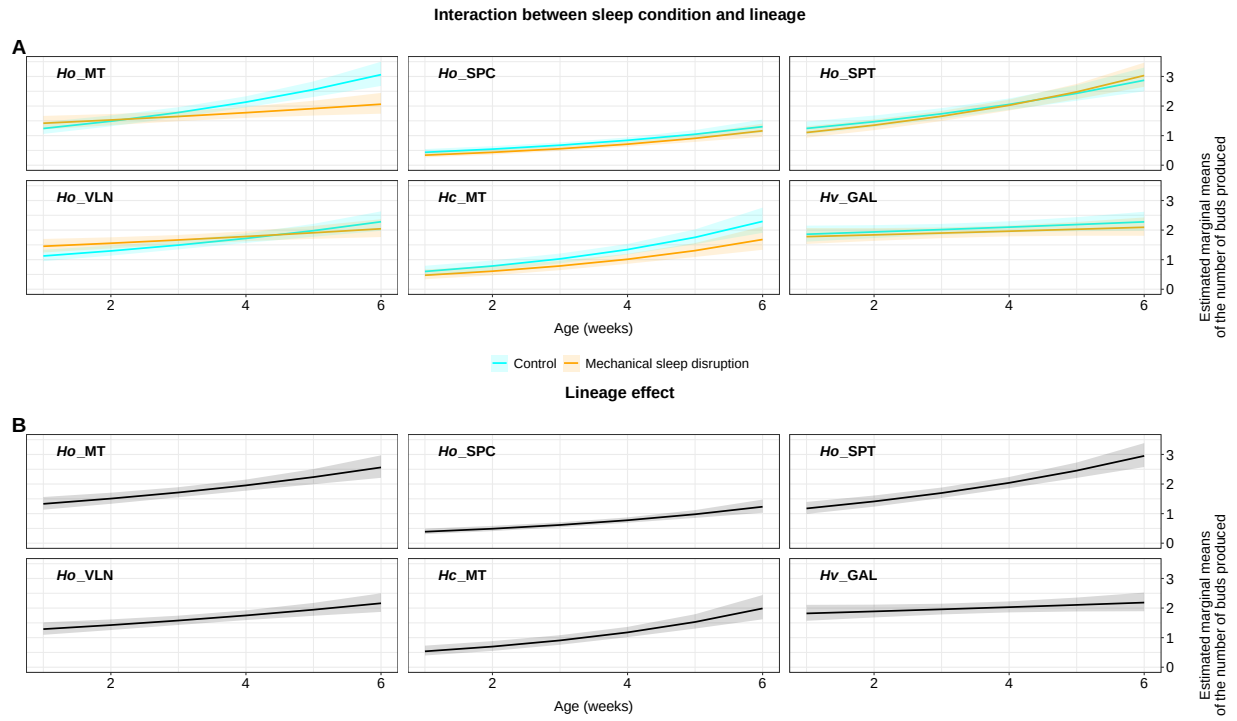
```
interaction_title_grob <- cowplot::ggdraw() +  
  cowplot::draw_label("Interaction between sleep condition and lineage",  
                      fontface = "bold", size = 16, x = 0.5, hjust = 0.5)  
  
lineage_title_grob <- cowplot::ggdraw() +  
  cowplot::draw_label("Lineage effect",  
                      fontface = "bold", size = 16, x = 0.5, hjust = 0.5)
```

Stack the titles, panels, and legend vertically into a single column.

```
final_buds_panel <- cowplot::plot_grid(interaction_title_grob,  
                                       A,  
                                       legend_treatment,  
                                       lineage_title_grob,  
                                       B,  
                                       ncol = 1,  
                                       rel_heights = c(0.08, 1, 0.1, 0.08, 1)) + #  
                                       Valeurs modifiées ici (0.08)  
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Show the final figure.

```
print(final_buds_panel)
```



```
ggsave("figure_5.png", plot = final_buds_panel, width = 21, height = 29.7, units = "cm",
dpi = 300)
```